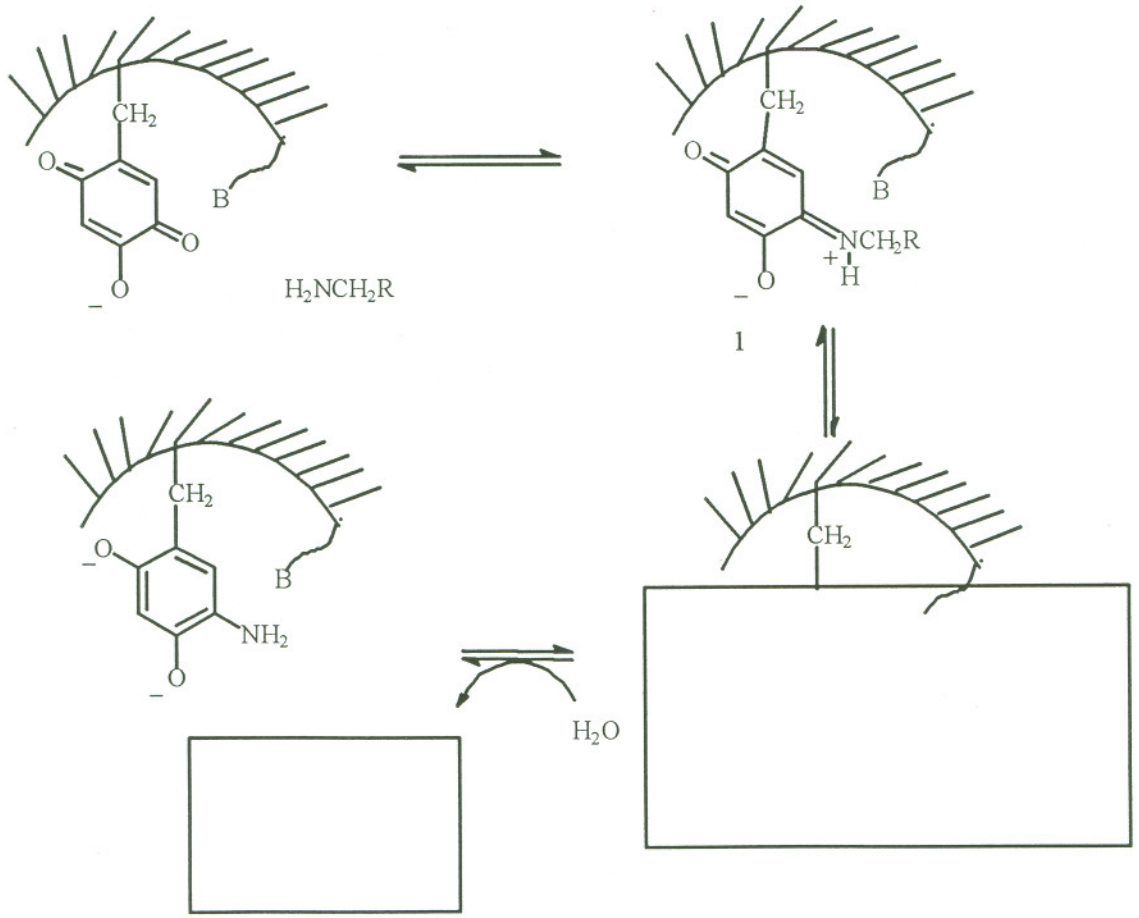
Organic Chemistry Cumulative Exam
September 19, 1992

Professy Judith Klinman recently gave a departmental seminar entitled " Quinoproteins: New Redox Cofactors in Enzyme Catalysis". Reference: Janes et al. *Science*, **1990**, *248*, 981-987. Using your knowledge of mechanistic and spectroscopic methods in organic chemistry, answer *three* of the following four questions.

1) Outlined below is the proposed chemical mechanism for the enzyme plasma amine oxidase. Fill in the missing intermediate structures in the boxes and use "arrow-pushing" to show the direction of electron flow.

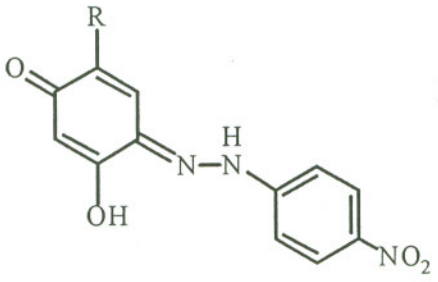
FIGURE A



2) When the intermediate **1** is reacted with $NaBT_4$, no radioactive tritium is found to coelute with the active-site peptide containing the cofactor When the intermediate **1** is generated using ^{14}C-labeled benzylamine, the radioactivity coelutes with the peptide. Explain these observations in terms of the proposed mechanism in Figure **A**.

3) Discuss the spectroscopic methods used to characterize the structure of the cofactor.

4) Figure **B** shows part of the 500 MHz ^1H NMR spectrum of the *p*-nitrophenylhydrazone derivative **2** in D_2O. The peak at 5.6 ppm slowly disappears with time. Assign all of the peaks in Figure **B**. Note: the small singlet peak at ~ 8.5 ppm is an impurity. **Discuss your reasoning.**



R = methylhydantion

2

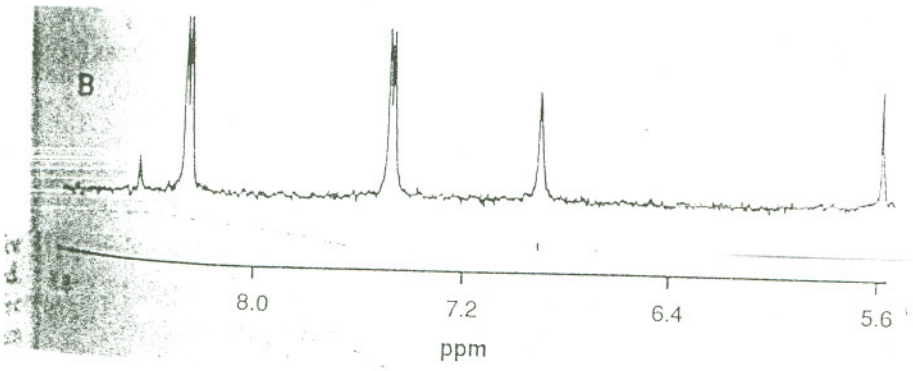



FIGURE B